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Problem

ATTENTION IS NOT FREE

- Modern diffusion backbones moved from CNN U-Nets to scalable DiTs, gaining quality but increasing latency.
- A plain 3 x 3 ConvNet is fast and hardware-friendly, yet its local receptive field limits generation quality.
- The challenge is to scale pure convolution without self-attention.

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Architecture

HOURLGLASS BEATS ISOTROPIC

- A basic residual block uses only two full stride-1 3 x 3 convolutions, GroupNorm and activation.
- Encoder downsampling expands effective receptive field; decoder upsampling restores spatial detail.
- Sparse, strided skip connections retain information without the cost of blockwise dense skips.

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Conditioning

MATCH EACH STAGE

- Each encoder/decoder stage gets its own timestep/class embedding aligned to its feature dimension.
- Injecting conditions in the middle of the block works better than only at the block boundary.
- AdaLN-style channel gates add dynamic control; replacing SiLU with GELU gives a final small gain.

DIC

Rethinking Conv3x3 Designs in Diffusion Models

A Pure Convolutional Diffusion Backbone | 2025

Recover the speed of hardware-optimized 3 x 3 convolution while matching transformer diffusion quality through an hourglass receptive field, sparse skips and

stage-aware conditioning.

Conv3x3

diffusion

U-Net

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Roadmap

ABLATION GAINS

- At 200K steps, isotropic Conv3x3 starts at FID 29.31; U-Net plus sparse skips reaches 11.49.
- Stage-specific embedding -> 10.07; mid-block injection -> 8.80; conditional gating -> 6.54.
- The final GELU-based DiC reaches FID 6.26 at essentially the same 116.1G FLOPs.

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Evidence

FAST AND COMPETITIVE

- At 400K, DiC-XL FID is 13.11 vs 20.05 for DiT-XL/2; throughput is 313.7 vs 66.8.
- With guidance DiC-XL reaches FID 3.89; DiC-H reaches FID 2.25 after 2M steps at 160.8 throughput.
- At 512 x 512, DiC-XL achieves FID 15.32 and 84.2 throughput vs DiT's 20.94 and 16.2.

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Meaning & limits

TAKEAWAY

- Simple convolutions remain highly competitive when macro-architecture and conditioning match their inductive bias.
- Linear spatial scaling and Winograd support make DiC attractive for higher resolution and real-time use.
- Evidence is mainly class-conditional ImageNet; text/video generation and broader comparisons remain future work.